Sentence Memorability Test - A study across the blocks

BRSM PROJECT POSTER PRESENTATION SP 2026

ABSTRACT

Statistical analysis was conducted on sentence memorability dataset to investigate the determinants of recall through a continuous recognition task focusing on Subject-Verb-Object (SVO) structures. While prior research has emphasized the role of mental imagery and reconstructive processes in sentence memory , this study specifically examines the effects of syntactic voice (A/P) and noun saliency (H/L).
We conducted analysis on corrected memorability scores(among participants), reaction times and hit rate across blocks. We have found significant differences in memorability scores across different conditions and fatigue effects displayed in higher RT as time passes.

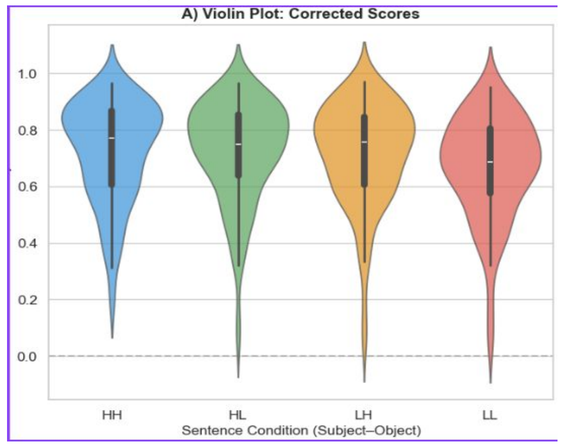
METHODOLOGY

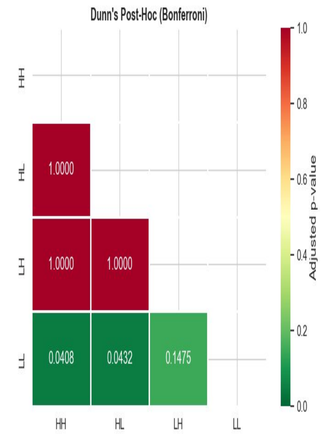
The main dependent variable was the corrected memorability score which takes care of response bias. Normality was tested using Shapiro-Wilk Test and as it showed non-normal, thus non-parametric methods were used.

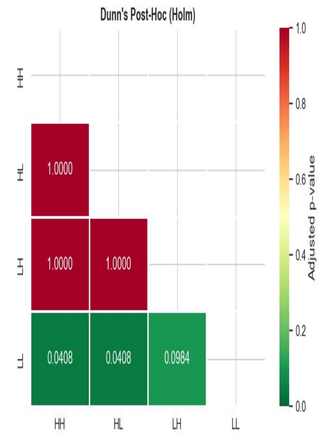
1. Kruskal-Wallis with Dunn's post-hoc (Holm correction) for memorability across conditions (HH, HL, LH, LL).
2. Wilcoxon Signed-Rank test for paired voice comparison (Active vs Passive).
3. Friedman test plus Page's trend test with paired Wilcoxon post-hoc (Holm correction) for block-wise accuracy trends.
4. Kruskal-Wallis plus Dunn's post-hoc (Holm correction) for reaction time differences

RESULTS

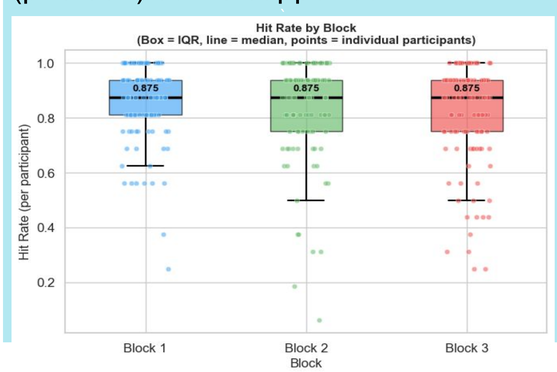
H1-Noun Memorability since data was non-normal thus Kruskal-Wallis test showed a significant effect $H(3)=10.00$, $p=0.019$, $\eta^2=0.016$ and Post-hoc(Dunn's, Holm) showed LL significantly lower than HH and HL ($p=0.041$) also LH vs LL was very marginal, observed was that HL ≈ HH > LH > LL, showing a threshold effect.
H2-Voice Effect where we did Wilcoxon Signed-Rank test showed no significant difference ($W=1612$, $p=0.200$, $r=0.158$) in which Active (0.702) and Passive (0.713) were similar.



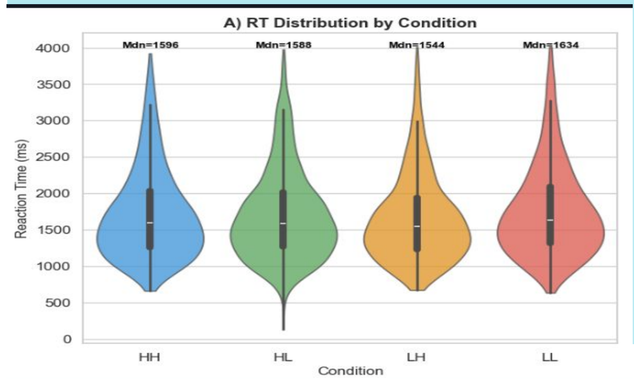




H3-Block/Fatigue Effect since Non-normal data thus Friedman test significant ($\chi^2(2)=7.81$, $p=0.020$) and Page's L test ($p=0.010$) confirm a monotonic decline with Wilcoxon (Holm) shows only B1 vs B3 significant ($p=0.013$), with Spearman marginal ($p=0.052$) so H3 supported.



LH4-Reaction Time Differences showed non-normality with Kruskal-Wallis test significant $H(3)=14.87$, $p=0.002$, $\eta^2=0.003$ and Dunn's post-hoc (Holm correction) and only LH vs LL significant ($p=0.001$, $r=0.096$) but only partial support



CONCLUSION

From the results we see that having a high noun in subject can significantly improve memorability scores however the voice has no effect in memorability unlike how it effects reconstruction. Also a fatigue effect is observed across blocks as RT increases and hit rate decreases.

CONTRIBUTIONS

All teammates have worked equally
Arnav - Noun memorability& group differences
Nidhish - RT hypo , literature & Transformations check
Nilavra -Voice Effect & Visualization

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Author names: Arnav Deshmukh, Nidhish Jain, Nilavra Ghosh (Team p < 1)
email ID: arnav.deshmukh@students.iiit.ac.in